

Raspberry pi, Pi camera, Motor driver, ultrasonic sensor module, raspbian OS, python, OpenCV.

According to this study, [5] there are two sub-systems that are used to build this project. That is Image processing sub-system and obstacle detection subsystem. Camera is attached to image processing sub-system which captures the image. To detect the road lane mainly image processing is used here. To detect the obstacle in front of car and also calculate the distance between the obstacle and the car, obstacle detection sub-system is used. These are captured with help of a webcam interfaced to the Raspberry Pi. It is convenient to use Raspberry Pi as it has very less software glitches and provides overall performance. System takes out the data from the image and generates the command about turn. Generated commands are forward to obstacle detection subsystem. If adequate distance is available to move car forward the command from Raspberry pi is forwarded to motor driver else this command are rejected. It is significant to detect the lane from the image of road and check their position in the form of pixel coordinates for the settlement of turn. To perform the lane detection action it is significant to convert image into gray scale. The white pixels from the region of interest are considered only. Position of lane on the road is given by the count of this image. The region of interest is slide vertically with respect to speed of car. Ultrasonic module provides the distance between two objects. Arduino's serial communication gives us stop command to the vehicle and also results about the distance. In this project, the main aim is at the development of self-driving vehicles for convenient transportation without a driver. Cars that drive themselves will improve road safety,

accessibility, fuel efficiency and increase productivity.

METHODOLOGY

The proposed project design mainly focuses on developing autonomous self-driven RC car. We do this by developing an RC car with a RPI board, camera and other multiple sensors. A basic RC car is assembled with DC motors, which receives the inputs from RPI board. The image of the road is taken from the RPI camera to the board via USB port.

The camera sends high quality video to the board which acts as input to the machine learning algorithms which decides the movements of the RC car.

There are three main divisions to bring about this: lane detection algorithm, convolutional neural network and ultrasonic sensors. Signs and ultrasonic sensors which give us the distance between the obstacles, we decide the next movement of the RC car.

A threshold distance is set between the obstacles and the car as 20cm. If the distance is below 20cms, the car stops no matter what. If the distance is more than 20cms and the CNN algorithm recognizes a stop sign, the RC car comes to halt. If both the outputs pass the threshold value, then the lane detection algorithm gives the RC car the steering angle. To make a left turn if necessary, the output voltage for the left server motor is reduced. To make a right turn if necessary, the output voltage for the right server motor is reduced. To make a halt, we stop the output voltage for both server motor.

The algorithm used to detect stop sign is HAAR cascade. It is a classifier which detects the object that it has trained for. It is done by superimposing positive image over a set of negative images.